

Samir Bhusal | Software QA Engineer | SDET

samir.masterprep@gmail.com | bhusalsamir.com.np | github.com/samirbhusal

Summary

Performance-driven Software QA Engineer / SDET with around 3 years of experience architecting scalable BDD test automation frameworks for regulated healthcare and biotech platforms. Proven expertise across Playwright, Selenium, Appium, Java, and TypeScript, building CI/CD-integrated regression suites in Azure DevOps that cut release cycle time by 30% and streamline QA across web, mobile, and API layers.

Education

Dakota State University | Madison, SD
Master of Science, Information Systems
Purbanchal University | Kathmandu, Nepal
Bachelor of Science, Computer Applications

August 2025 – Dec 2026

October 2017 – October 2022

Experience

Software QA Automation Intern | Bio-Techne, Minneapolis, MN

June 2026 – August 2026

- Architected a multi-environment (Dev/QA/UAT) Playwright + TypeScript + Playwright-BDD framework using Page Object Model.
- Built Azure DevOps pipelines — smoke/PR validation and nightly regression — with PowerShell-triggered execution from the Azure Classic Release Pipeline to run test suites automatically on deployment completion.
- Integrated Allure Report dashboards for stakeholder-facing test visibility.
- Synced automated test outcomes with Azure DevOps Test Plans in real time, replicating Xray-style traceability natively.
- Delivered a Playwright framework walkthrough to the full Bio-Techne IT organization (Denver + MN sites) and an intern capstone presentation.
- Built custom Claude Code skills using Playwright MCP to auto-draft Gherkin test scenarios from the live UI, and evaluated no-code alternatives (Katalon Studio, BrowserStack LCA, Leapwork).

Graduate Research Assistant | Dakota State University, Madison, SD

August 2025 – May 2026

- Designed AI helpdesk assistant **prototype** for incident-resolution workflows.
- Supported XAI healthcare research by maintaining literature matrix and evaluating Python analytics projects.
- Assisted in rubric design and grading for analytics coursework.

Software Engineer – Quality Assurance | Deerhold Ltd., Kathmandu, Nepal

November 2022 – August 2025

- Maintained BDD automation framework for **Higi platforms** (web + mobile) using Selenium, Appium, and Java, achieving 95% regression automation coverage, 90%+ pass rate, and reducing release cycle time by 30%.
- Designed and maintained BDD test automation framework for **InStride Health** (mobile app – Android & iOS)) with 80% automation coverage using Selenium, Appium and Java.
- Executed API tests to prepare test data and preconditions for web and mobile automation execution.
- Led QA process for legacy system rebuilds, achieving production readiness in 3 months for **Wellness Portal**.
- Managed test planning, test case design, execution, defect tracking, and release documentation across web and mobile platforms.
- Partnered in agile ceremonies ensuring alignment with cross-functional teams for continuous improvement.

Projects

Python BDD Automation Framework | Python, Behave, Selenium, Page Object Model

- Developed a scalable Python Behave + Selenium POM framework with multi-browser & environment-driven execution.

Playwright BDD Automation Framework | TypeScript, Playwright, Cucumber

- Developed a scalable Playwright BDD automation framework with UI/API testing, multi-browser execution, BrowserStack integration, and Allure reporting.

Full Stack RAG Chatbot | React, Python, FastAPI, LangChain, Pinecone, Groq (Llama-3.3), HuggingFace

- Developed a RAG chatbot for intelligent incident query resolution using internal documents.

Technical Skills

Languages:	Java, Python, TypeScript, SQL, Gherkin
Test Automation:	Playwright, Playwright-BDD, Selenium, Appium, Cucumber, Behave
Tools & Platforms:	Git, GitHub Actions, Azure DevOps (Pipelines & Test Plans), JIRA, BrowserStack, Postman, Allure Report
Testing:	API, Regression, Smoke, UAT, Cross-Browser / Cross-Platform
Methodologies:	Agile/Scrum, SDLC, STLC, CI/CD

Achievements / Awards

- Winner of Hult Prize Purbanchal University 2020